

Sanjay J K

Software Engineer — AI Systems Developer
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SUMMARY

Computer Science student at VIT (CGPA 9.34) building scalable full-stack software systems, from event-driven backend architectures to production-grade web applications, with applied AI/ML components (LLM pipelines, RAG) where they add real value. Comfortable across the stack: API design, databases, and system architecture. Holds two granted patents in applied AI.

SKILLS

Systems & Backend: Event-Driven Architecture, REST APIs, WebSockets, OAuth2, Asynchronous Job Processing, Data Structures & Algorithms, Docker, Redis, PostgreSQL, MongoDB, SQLite, SQL, ChromaDB, Neo4j

AI / ML: LLM Pipelines, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, PyTorch, Scikit-learn, spaCy, NLP, Model Deployment

Languages: Python, TypeScript, JavaScript, C++, Java, C

Frameworks & Web: React, Next.js, FastAPI, Node.js, Express.js, TailwindCSS, Zustand

Tools: Git, GitHub, Pytest, LangChain, LangGraph, Langfuse, Ollama, Docker Compose

PROJECTS

LLM Knowledge Assistant (Production RAG System) Python, LangChain, ChromaDB

- Engineered a **chunking and embedding pipeline** (windowed segmentation with overlap, MiniLM embeddings) feeding a **hybrid retrieval system** that merges dense vector search and BM25 lexical search via Reciprocal Rank Fusion.
- Improved answer precision with **cross-encoder re-ranking** and a citation-confidence threshold that declines low-relevance queries; validated pipeline quality via a golden-dataset evaluation suite with full Langfuse tracing.

InterviewAI — Real-Time Voice AI Interviewer FastAPI, React, WebSockets, Ollama

- Built a real-time voice AI interviewer (Deepgram ASR, Ollama LLMs, ElevenLabs TTS) achieving **sub-1.2s end-to-end latency** via an event-driven, full-duplex WebSocket streaming architecture.
- Engineered a **zero-lag interruption handler** that cancels in-flight LLM/TTS generation and flushes client audio buffers mid-response, enabling natural conversational turn-taking.

NextUp — AI-Powered Placement Intelligence Platform Next.js, FastAPI, PostgreSQL, APScheduler

- Architected a **zero-knowledge encryption system**: student data is encrypted client-side (AES-256-GCM), with the backend storing ciphertext only and keys never persisted server-side.
- Built an automated **Gmail-to-workflow pipeline** (NLP-based email parsing, fuzzy-matched shortlist detection) feeding interactive Kanban/calendar dashboards and a client-side resume parser for auto-populated profiles.

Neo4j GraphReasoner — NLP Knowledge Graph Neo4j, Node.js, Ollama

- Built a **knowledge graph pipeline** ingesting real-world research data (OpenAlex API) into Neo4j, implementing 10+ Cypher queries for multi-hop reasoning, structural inference, and citation-centrality analysis.
- Developed an **LLM-to-Cypher query engine** (Ollama) with a semantic auto-correction layer preventing hallucinated graph properties, plus end-to-end observability (Langfuse tracing, structured logging, threshold-based alerting).

PATENTS

- Published (2026, Indian Patent Office):** Personalized AI System for Color Vision Deficiency — an adaptive recommendation engine using perceptual gamut modeling to assist colorblind users.
- Published (2026, Indian Patent Office):** Pan-Chronological Vision-Language Transformer for Cross-Era Tamil Script Decipherment — a VLM system for reading historical Tamil inscriptions across centuries of script drift.

EXPERIENCE

International Research Scholar — UKIERI-SPARC Program, Heriot-Watt University Jun 2025

- Selected as **1 of 9 students nationally** for an international hydrogen-energy research exchange; applied iterative, data-driven experimentation (cyclic & linear sweep voltammetry) to catalyst validation, drawing on the same analysis-and-abstraction habits used in debugging and systems work.
- Co-presented a research-to-startup roadmap (technical findings, budget, and scalability plan) to a faculty panel, translating experimental data into a structured business case.

Frontend Developer Intern — Valsco Technology (Remote) Jun 2024 – Sep 2024

- Scaled **NoteSwap** to **200+ active student users** by architecting a component-based React frontend with reusable UI modules and REST API integration, cutting page load time by **20%** through lazy loading and render optimization.
- Improved cross-device usability by designing a responsive component library adopted across the full desktop and mobile interface.

EDUCATION

Vellore Institute of Technology (VIT) — B.Tech Computer Science 2023 – 2027

CGPA: **9.34**

CBSE High School Score: **93.5%**

RECOGNITION

- GATE CSE 2026: AIR 1603** (Score: **670**).
- Judges' Choice Award – Sankalp Innovation Challenge 2026 (MNNIT Allahabad):** Built **JanVedha AI**, a civic governance platform combining grievance analytics, explainable AI prioritization (LightGBM/SHAP), and a Gemini-powered decision assistant for local officials.

HACKATHONS & LEADERSHIP

- Finalist – Google Cloud Agentic AI Hackathon 2025; Finalist – India Innovates 2026 (Bharat Mandapam); Top 3 – DeepShiva Open Source LLM Hackathon.
- Board Member (Design Lead), AEE VIT & Senior Core Member, CodeChef VIT / TAM-VIT.